dot(dot(X, Y), Z) → dot(X, dot(Y, Z))

T = dot(X, Y), O = dot(T, Z). Inner dot contracts K; outer dot contracts L.

1. Select axes

I J

T: i_0 i_1 j_0 j_1

0 1 2 3

c_L = [3, 0]

L: j_1 i_0

slot 0 slot 1

Tensor order is unchanged

→

2. Partition by origin

I = [i_0, i_1]

I_sel [i_0] I_rem [i_1]

v_I = [0], u_I = [1]

J = [j_0, j_1]

J_sel [j_1] J_rem [j_0]

v_J = [1], u_J = [0]

Views retain axis positions

→

3. Preserve pairings

Physical Z axes

Z: z_0 f z_1

0 1 2

c_R = [2, 0]

Slot	From T		From Z	Z pos
0	j_1	↔	z_1	2
1	i_0	↔	z_0	0

Selected subgroup mappings

Positions are zero-based

Selected group	Producer positions	Slots in L	Partner positions in Z
I_sel	v_I	u_I	gather(c_R, u_I)
J_sel	v_J	u_J	gather(c_R, u_J)

U = dot(Y, Z) contracts J_sel; final dot contracts I_sel + K.

U retains K and I_sel's partners. Output: I_rem ++ J_rem ++ F (F = [f]).